Project Proposal : Machine Learning for Finance Fraud Detection

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Overview

The application of machine learning in fraud detection is expected to gain momentum due to its ability to process large datasets and uncover hidden patterns indicative of fraudulent activities. **Logistic Regression**, combined with **Gradient-based Optimization** techniques, will offer a scalable and effective approach for identifying anomalies in financial transactions. These methods will be particularly well-suited for fraud detection, where binary classification is essential to distinguish between fraudulent and legitimate transactions.

Logistic Regression will serve as a foundational model for binary classification, making it a natural fit for fraud detection tasks. This algorithm will estimate the probability of a transaction being fraudulent based on input features. Its interpretability will allow stakeholders to understand the key factors contributing to fraud detection decisions, making it a strong starting point for financial institutions. However, Logistic Regression's linear nature may limit its ability to capture more complex, non-linear relationships in the data. This limitation will be addressed by incorporating more advanced optimization techniques and regularization methods.

To improve the performance of Logistic Regression, especially in high-dimensional and imbalanced datasets, **Gradient-based Optimization** methods, such as **Gradient Descent**, will be employed. Gradient Descent will help in finding the optimal parameters that minimize the cost function, typically binary cross-entropy for classification problems. **Stochastic Gradient Descent (SGD)** and **Mini-batch Gradient Descent** will further enhance this process by making it computationally efficient and better suited for large-scale datasets, where fraud detection will often be applied.

A major challenge in fraud detection will be the **class imbalance**, where fraudulent transactions are much rarer than legitimate ones. This imbalance may lead Logistic Regression to favor the majority class (legitimate transactions). To mitigate this, techniques like **SMOTE (Synthetic Minority Over-sampling Technique)** and **cost-sensitive learning** will be used to balance the dataset or adjust the model's focus towards the minority class. Additionally, regularization techniques like **L1 (Lasso)** and **L2 (Ridge)** will be applied to prevent overfitting and improve the model's generalization capability.

By leveraging **Logistic Regression** optimized through **Gradient-based methods**, this project aims to build a robust fraud detection system. Although Logistic Regression will be relatively simple, its effectiveness will be enhanced when combined with feature engineering, regularization, and class balancing strategies. The final model will be capable of detecting fraudulent transactions in real time, offering financial institutions a scalable and transparent solution for reducing financial losses due to fraud.

Preliminary Approach

The approach for detecting fraudulent activities will begin with **exploratory data analysis (EDA)** to understand the data distribution, identify potential features, and address data quality issues such as imbalance. This step will be crucial for recognizing the relationships between features and determining how best to prepare the data for **Logistic Regression**. Special focus will be placed on handling **class imbalance**, as fraudulent transactions will typically be much less frequent than legitimate ones.

Data balancing techniques such as **SMOTE** and **cost-sensitive learning** will be applied to ensure the model can adequately learn to classify both legitimate and fraudulent transactions. In parallel, **feature engineering** will be used to create relevant variables that help improve the model's predictive accuracy. Feature selection techniques will be considered to enhance the efficiency of the Logistic Regression model by keeping only the most informative features.

The model will be trained using **Logistic Regression** as the core algorithm, optimized through **Gradient-based methods** like **Stochastic Gradient Descent (SGD)**. Regularization techniques, such as **L1 and L2**, will be employed to handle potential overfitting. The use of **Gradient-based optimization** will allow the model to iteratively update its parameters to minimize the cost function, leading to more accurate predictions.

Logistic Regression and Gradient Descent

Logistic Regression will be the primary method used to predict the likelihood of a transaction being fraudulent. It will be optimized using **Gradient Descent**, which will iteratively adjust the model's parameters to minimize the error in classification. **Stochastic Gradient Descent (SGD)** will be used to make the optimization process more efficient, particularly for large datasets, by updating parameters on a per-sample basis.

Given the class imbalance challenge, strategies like **SMOTE** and **cost-sensitive learning** will ensure that the model can detect fraud effectively without being biased toward the majority class. **Regularization** techniques such as **Lasso** and **Ridge** will also be integrated to avoid overfitting the model to the training data, further enhancing its generalizability.

Dataset

- **Credit Card Fraud Detection Dataset**: The dataset contains transactions made by credit cards in September 2013 by European cardholders. This dataset will be used for training and evaluating the performance of the Logistic Regression model in detecting real-world fraudulent transactions. It will be highly imbalanced, which reflects the typical challenges faced in fraud detection. Key features will include transaction amount, time, and anonymized variables generated from principal component analysis (PCA).
  - Dataset Link

Evaluation Metrics

The evaluation of the **Logistic Regression** model will rely on several metrics, particularly chosen to handle the **imbalanced nature** of fraud detection:

- **Accuracy**: While accuracy will be a basic metric, it may be misleading in imbalanced datasets, where the majority class dominates.
- **Precision, Recall, and F1-Score**: Precision will measure the proportion of transactions predicted as fraudulent that are truly fraudulent. Recall will assess how many actual fraudulent transactions will be correctly identified by the model. The F1-score will provide a balance between precision and recall, making it ideal for handling imbalanced datasets.
- **Area Under the ROC Curve (AUC-ROC)**: This metric will be used to evaluate how well the model distinguishes between fraudulent and legitimate transactions, providing an aggregate performance

measure across various thresholds. The AUC-ROC curve will be plotted with a **True Positive Rate (TPR)** against a **False Positive Rate (FPR)**, where:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{1}$$

where TP will be True Positives, FP will be False Positives, TN will be True Negatives, and FN will be False Negatives.

Task Breakdown and Milestones

1. **Week 1: Literature Review**
    - Review recent studies on fraud detection using Logistic Regression and Gradient Descent.
    - Identify techniques for addressing class imbalance and feature engineering.
2. **Week 2-3: Data Exploration and Preprocessing**
    - Conduct EDA to understand the dataset structure and identify important features.
    - Apply SMOTE and cost-sensitive learning to address data imbalance.
    - Perform feature engineering to enhance model performance.
3. **Week 4-5: Model Implementation**
    - Implement **Logistic Regression** using **Gradient-based optimization**.
    - Integrate regularization techniques (**L1, L2**) to prevent overfitting.
    - Evaluate model performance using the chosen metrics.
4. **Week 6: Model Evaluation and Final Report**
    - Analyze false positives and false negatives to identify areas for improvement.
    - Fine-tune hyperparameters using grid search to improve model performance.
    - Compile findings and prepare a detailed report.
5. **Roles and Responsibilities**
    - **Krishna Mattaparthi**: Lead for data exploration, preprocessing, and handling class imbalance techniques. Responsible for model implementation, including integrating Logistic Regression and Gradient Descent.
    - **Justin Jiang**: Lead for literature review and feature engineering. Responsible for implementing evaluation metrics, focusing on the AUC-ROC and performance analysis.
    - **Hemansh Adunoor**: Support for model training and optimization, particularly with Gradient Descent and regularization techniques. Assist with fine-tuning the model.
    - **Kyoungnam Moon**: Responsible for creating the final report, analyzing model performance, and coordinating overall progress. Assist with model evaluation and preparing the final report.

Summary

This project aims to utilize **Logistic Regression** optimized through **Gradient-based methods** for financial fraud detection. Addressing class imbalance through SMOTE and cost-sensitive learning, the project seeks to develop a robust, scalable solution for identifying fraudulent transactions. The model's simplicity and interpretability, combined with regularization techniques and feature engineering, will provide an effective tool for real-time fraud detection in the financial sector.

References

- Financial Fraud Detection Based on Machine Learning: A Systematic Literature Review
- Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances
- Machine Learning for Financial Fraud Detection: A Comprehensive Review